AI-Powered Digital Marketplace for Upcycling Sri Lanka's Textile Waste

BSc (Hons) in Information Technology
Sri Lanka Institute of Information Technology

Project ID: R26-IT-081



SLIIT
FACULTY OF COMPUTING

Our Team



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Project Supervisor



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Project Co-Supervisor

Our Team

Meet the dedicated individuals driving this innovative project:



Sandeepa S.A.C.

AI Fabric Classification Module



Siriwardhana N.S.T.N.

Environmental Impact Prediction Module



Madushanka H.M.T.G.

Traceability, Recommendation & AI Pricing Module



Thalagala T.A.D.V.

Marketplace Platform & Predictive Analytics Module

INTRODUCTION

Introduction to the problem

The main research problem is the lack of an intelligent and integrated digital system for textile waste management in Sri Lanka’s apparel industry. Currently, textile waste is manually sorted and often burned or disposed of in landfills, causing environmental pollution and economic loss. Manufacturers face difficulties in managing and valuing waste materials, while designers, upcyclers, and SMEs struggle to find affordable and reliable reusable fabrics.

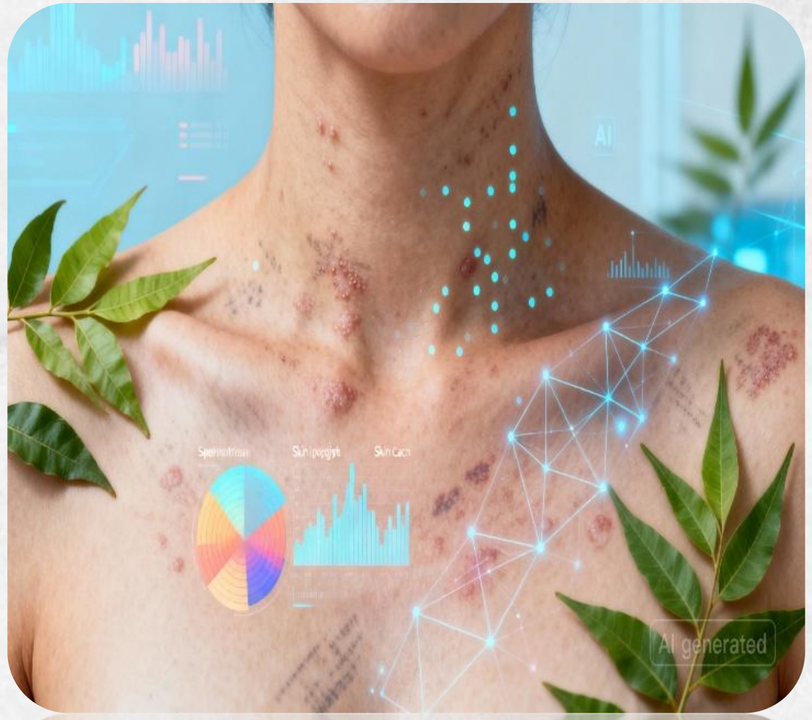
Social or research impact of the solution

To address these challenges, this research focuses on developing an AI-powered marketplace that enables automated fabric classification, environmental impact analysis, intelligent pricing, traceability, and predictive analytics to support a sustainable circular economy.



Our Approach to Address Gaps

- AI-powered marketplace
- CNN fabric classification
- Environmental impact prediction
- Microservices architecture Traceability system
- AI pricing & recommendation
- Predictive analytics dashboard



Constraints and Limitations

- Limited availability of Sri Lankan textile waste datasets
- Prediction accuracy depends on image quality and available data
- Environmental impact calculations use secondary LCA data sources
- Limited historical transaction data for AI pricing models
- Time and resource limitations for large-scale deployment and testing
- User adoption may vary based on awareness and technology usage skills

RESEARCH QUESTION

How can an AI-powered digital marketplace improve textile waste classification, environmental impact prediction, pricing, traceability, and waste redistribution in Sri Lanka?



RESEARCH GAP

Existing dermatological AI models rely heavily on single-modality image data and provide limited transparency in their decision-making. Many ignore relevant clinical metadata or diagnostic reports, reducing diagnostic accuracy and trustworthiness. Moreover, there is minimal work on underrepresented conditions like tinea and leprosy.

AI-Powered Textile Waste Management Platform – System Overview

Intelligent Analysis • Sustainability Prediction • Optimization • Marketplace Action

1 FABRIC INTELLIGENCE MODULE (Member 1 – Sandeepa)

Core Responsibility
Fabric Inspection & Condition Intelligence System

Key Features
- Fabric Type Classification (Cotton, Polyester, etc.)
- Color Detection
- Quality Grade Prediction
- Damage Detection (tear, stain, defects)
- Fabric Health Score (0–100)
- Repairability Prediction (repair / not worth)
- Lifespan Estimation (remaining usable time)

Example Output



Health Score: 78 / 100

Prediction: Repairable ✓
Lifespan Left: 6 – 12 months

Technologies
- CNN with Transfer Learning (ResNet / EfficientNet)
- Optional: YOLO for defect detection

2 SUSTAINABILITY INTELLIGENCE MODULE (Member 2 – Siriwardhana)

Core Responsibility
Environmental Impact & Decision Intelligence System

Key Features
- CO_2 Saving Prediction
- Water Saving Prediction
- Green Score
- Scenario Simulation (reuse vs recycle vs discard)
- Sustainability Score (0–100)
- Best Action Recommendation
- Trend Learning from historical data

Scenario Simulation
- Reuse: CO_2: -12.5 kg, Water: -250 L
- Recycle: CO_2: -8.2 kg, Water: -120 L
- Discard: CO_2: +2.4 kg, Water: +30 L

Sustainability Score: 86 / 100

Technologies
- Random Forest / XGBoost
- Feature Engineering and Simulation Logic

3 TRACEABILITY, RECOMMENDATION & AI PRICING MODULE (Member 3 – Madushanka)

Core Responsibility
Smart Value & Traceability Engine

Key Features
- AI Pricing Model
- Recommendation System (buyers / reuse options)
- Digital Material Passport
- Transaction Traceability
- Demand-aware pricing
- Context-aware recommendations

Material Passport (Example)



Material: Cotton
Color: Blue
Quality Grade: A
Health Score: 78/100
Lifespan Left: 6–12 m
Origin: Factory A
Date: 2024-05-18

Price Prediction
Estimated Price
LKR 320.00
(confidence: 92%)

Technologies
- Machine Learning (Pricing Model)
- Recommendation Systems
- Blockchain / Database for Traceability

4 MARKETPLACE & OPTIMIZATION INTELLIGENCE MODULE (Member 4 – Thalagala)

Core Responsibility
Marketplace Intelligence & Optimization System

Key Features
- Marketplace Platform (React UI)
- Waste Forecasting
- Smart Matching Engine (buyer-seller matching)
- Distance Optimization
- Cost Minimization
- Environmental Benefit Maximization
- Optimization using Weighted Scoring / Greedy Logic

Matching Overview



Distance: 18 km
Cost: LKR 3,200
CO_2 Saved: 14.5 kg

Technologies
- React (Marketplace UI)
- ML for Forecasting
- Optimization (Greedy / Weighted Scoring)

OVERALL SYSTEM FLOW

1. **FABRIC INPUT** — Fabric Image Collection
2. **CV ANALYSIS** — Fabric Intelligence (CV Model), Condition & Quality Assessment
3. **SUSTAINABILITY PREDICTION** — Environmental Impact Prediction & Scenario Simulation, Best Action Recommendation
4. **OPTIMIZATION DECISION** — Best Matching, Cost & Distance Optimization, Max Environmental Benefit
5. **MARKETPLACE ACTION** — List / Trade / Sell Transaction Completion

DATA & MODEL FLOW (Cross-cutting)

Image & Sensor Data → Preprocessing & Feature Extraction → AI/ML Models (CV, ML, Optimization) → Predictions & Insights → Database & Storage

STAKEHOLDERS

Factories (Sellers), Collectors, Recyclers (Buyers), Logistics Providers, Environment & Community

KEY OUTCOMES

- ✓ Reduce Textile Waste & Landfill
- ✓ Lower CO_2 & Water Footprint
- ✓ Increase Material Value Recovery
- ✓ Data-driven Sustainable Decisions
- ✓ Efficient Marketplace & Supply Chain

A sustainable, intelligent, and data-driven ecosystem that transforms textile waste into value while protecting the environment.

Tools and Technologies

- React.js
- Tailwind CSS
- FastAPI
- MongoDB Atlas
- Axios
- Python
- Scikit-learn
- Random Forest Algorithm
- Git & GitHub
- Vite
- Postman







Made with GAMMA

Sandeepa S.A.C.
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AI-Powered Fabric Inspection & Damage Detection System



INTRODUCTION & RESEARCH PROBLEM

INTRODUCTION

The Fabric Intelligence Module is an AI-based system developed to automate fabric inspection using Deep Learning and Computer Vision.

The system can:

- Identify fabric types
- Detect fabric damages
- Classify fabrics as Good or Defective

✓ Improves textile quality inspection accuracy

✓ Reduces manual effort

RESEARCH PROBLEM

Current Issues

- Manual inspection is slow and inconsistent
- Human errors reduce accuracy
- Difficult to identify small defects

Need for an AI Solution

- Faster inspection
- Accurate damage detection
- Real-time fabric analysis

RESEARCH OBJECTIVES & SYSTEM OVERVIEW

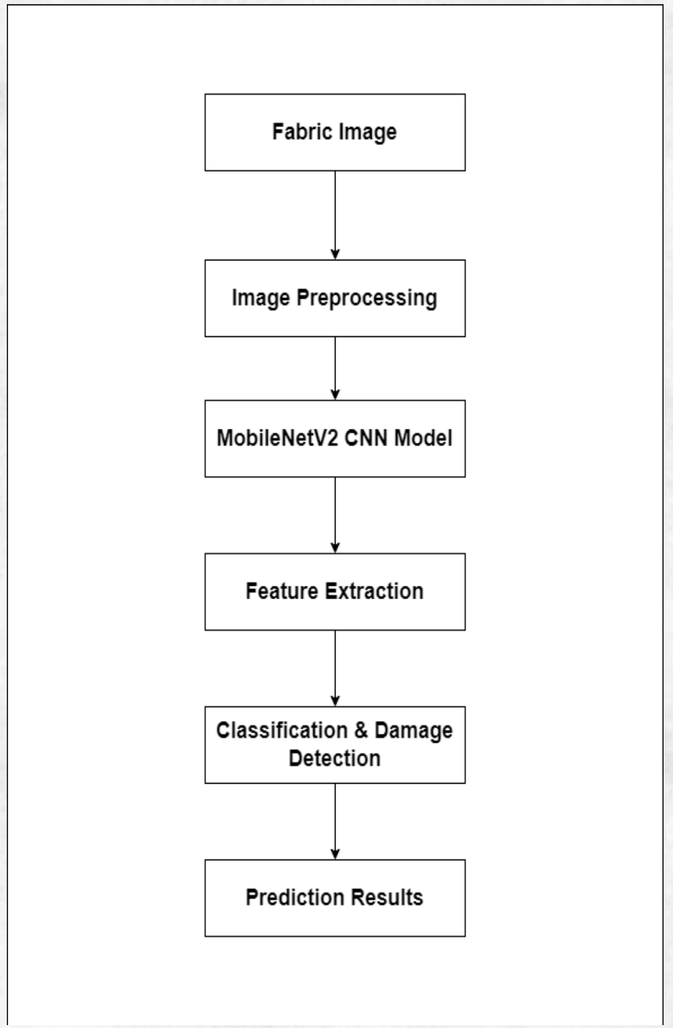
MAIN OBJECTIVE

Develop an AI-powered fabric inspection system using CNN models.

SPECIFIC OBJECTIVES

- Fabric type classification
- Damage detection
- Good vs Defective prediction
- Improve inspection accuracy

SYSTEM OVERVIEW



Outputs:

- Fabric Type
- Good Fabric / Defective Fabric

AI MODEL & TRAINING PROCESS

AI MODEL USED – MobileNetV2 CNN

Pretrained MobileNetV2 with Transfer Learning

Why MobileNetV2?

- High accuracy
- Fast processing
- Lightweight architecture
- Suitable for image classification

TRAINING PROCESS

Step	Description
1	Dataset Collection
2	Image Preprocessing
3	Data Augmentation
4	Transfer Learning
5	Fine-Tuning

Technologies: PyTorch, OpenCV, Python, Google Colab

FABRIC CLASSIFICATION & DAMAGE DETECTION

FABRIC TYPE CLASSIFICATION

Supported Types:

- Acrylic, Artificial leather, Blended, Chenille, Corduroy, Cotton, Crepe, Denim, Fleece, Felt, Leather, Linen, Nylon, Polyester, Satin, Silk, Suede, Terrycloth, Velvet, Viscose, Wool

Workflow:

Fabric Image → CNN Feature Extraction → Fabric Type Prediction

DAMAGE DETECTION

Detected Defects:

- Good Fabric
- Defective Fabric

KEY FEATURES & NOVELTY FEATURES

KEY FEATURES

- Fabric Type Classification – Automatically identifies different fabric materials
- Damage Detection – Detects visible fabric defects automatically
- Good vs Defective Prediction – Classifies fabric conditions intelligently
- Statistical Analysis – Provides model accuracy and condition analysis

NOVELTY FEATURES

- AI-Powered Automated Inspection (replaces manual methods)
- MobileNetV2 Transfer Learning (pretrained CNN for efficient learning)
- Intelligent Defect Identification (texture & pattern analysis)
- Real-Time Fabric Analysis (fast image-based predictions)

STATISTICAL ANALYSIS & CURRENT PROGRESS

STATISTICAL ANALYSIS

Feature	Accuracy
Fabric Classification	64.23%
Damage Detection	86.3%

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Dataset Statistics

Category	Percentage
Good Fabrics	68%
Defective Fabrics	32%

CURRENT PROGRESS

Status	Task
Completed	Dataset Collection
Completed	MobileNetV2 Integration
Completed	Fabric Classification
Completed	Damage Detection
In Progress	Statistical Analysis
In Progress	Design frontend

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❖ Sandler et al., *MobileNetV2: Inverted Residuals and Linear Bottlenecks*, 2018.

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❖ Islam, *CNN-Based Transfer Learning in Image Classification*, 2020.

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❖ Gondhalekar et al., *MobileNetV2 Approach for Image Classification*, 2025.

https://www.researchgate.net/publication/389687628_Enhancing_Image_Classification_Performance_through_Transfer_Learning_and_Adaptive_Augmentation_A_MobileNetV2_Approach

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Sustainability decision intelligence for textile waste routing, ML preparation, and marketplace integration.



INTRODUCTION & PROBLEM

What is the problem ?

- Textile waste is increasing rapidly.
- Difficult to decide whether items should be reused, repaired, recycled, or discarded.
- Manual decisions are inefficient and inaccurate.
- Poor waste management causes environmental pollution.
- Need for an AI-powered sustainability decision system.

Key Features

- Sustainability Assessment
- Scenario Simulation
- AI-Based Recommendation System
- Risk Assessment & Data Quality Analys
- Dashboard Analytics
- Explainable Recommendation Output
- Environmental Impact Prediction
- Integration Support for Pricing & Marketplace Modules

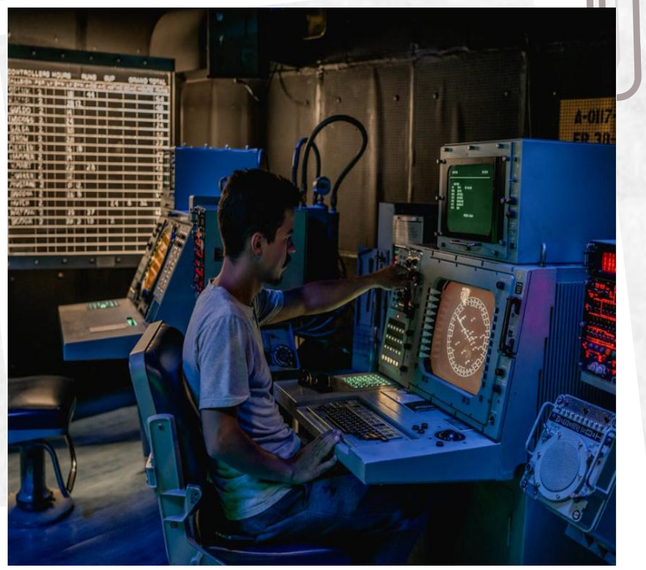
Research Gap

- Existing systems mainly focus on waste collection, not sustainability intelligence.
- Lack of AI-based decision support for textile waste management.
- Most platforms do not provide explainable recommendations or risk analysis.
- No integrated system for sustainability prediction and scenario comparison.
- Limited use of ML models for textile reuse and environmental impact evaluation.

Novelty Features

- Environmental impact prediction
- Scenario comparison across multiple disposal or recovery paths
- Explainable recommendation output
- Risk and data-quality assessment
- Integration payloads for pricing and marketplace matching components

Evaluation and Validation



- Regression model evaluated using MAE, RMSE, and R^2 Score.
- Classification model evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix
- API smoke testing used to verify backend functionality.
- Frontend and backend build verification completed successfully.
- Manual testing performed for prediction, scenario simulation, dashboard analytics, and ML workflow validation.
- System reliability validated through service and integration testing.

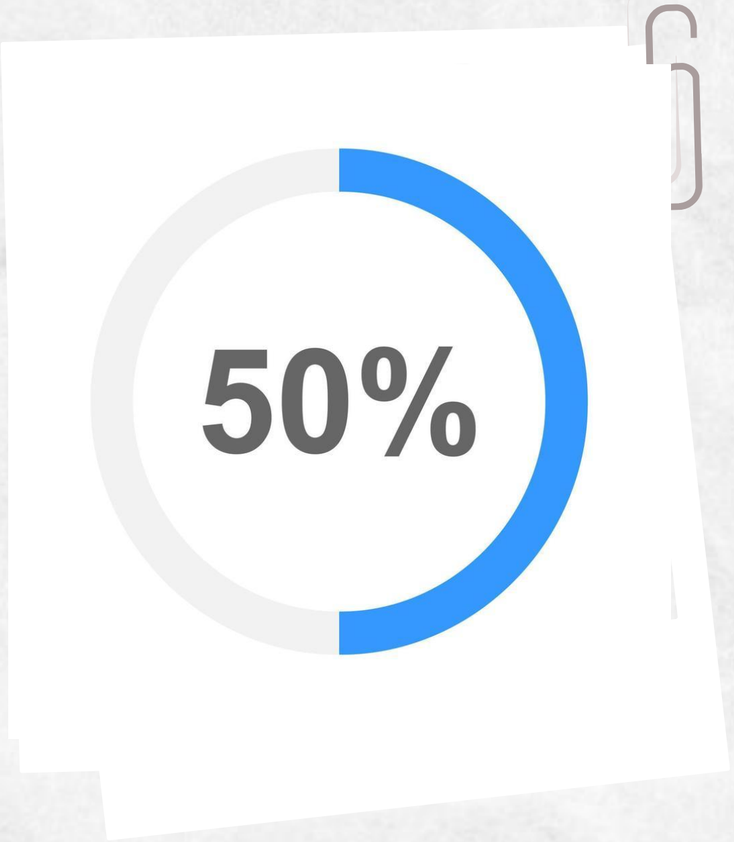
Data Availability & Ethical Clearance

- The current dataset is synthetic and generated for development and testing purposes.
- No personal or sensitive user data was collected.
- Data is used only for research and educational purposes.
- The system follows responsible AI and sustainability-focused practices.
- Future improvements will include real textile waste datasets with proper authorization and validation.



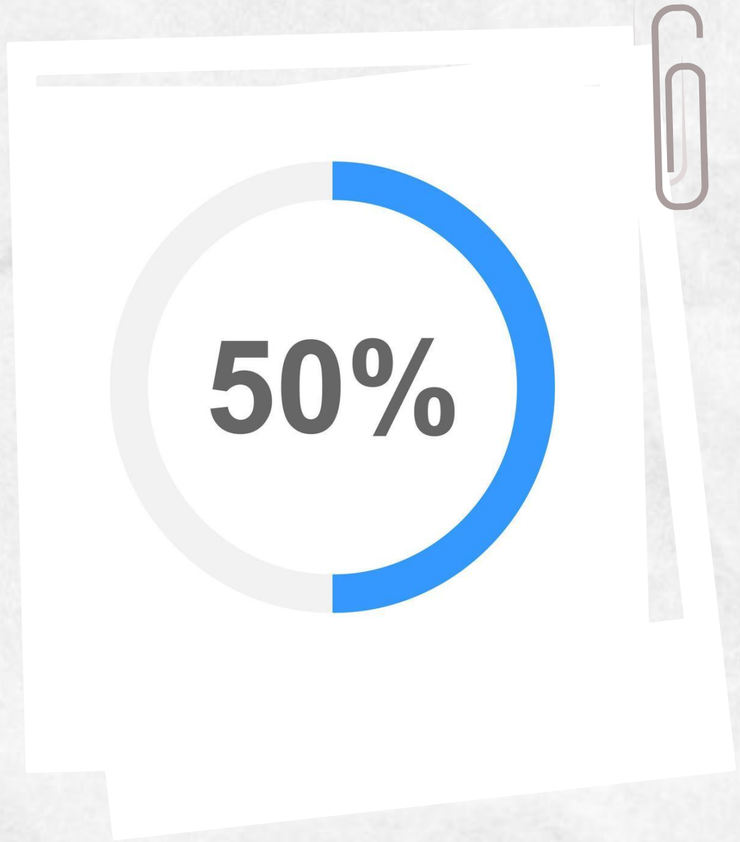
Current Progress 50 %

- Completed full-stack implementation using React.js, FastAPI, and MongoDB Atlas.
- Implemented sustainability prediction and scenario simulation features.
- Added risk assessment and explainable recommendation system.
- Developed dashboard analytics and prediction history modules.
- Completed frontend-backend integration and testing processes.
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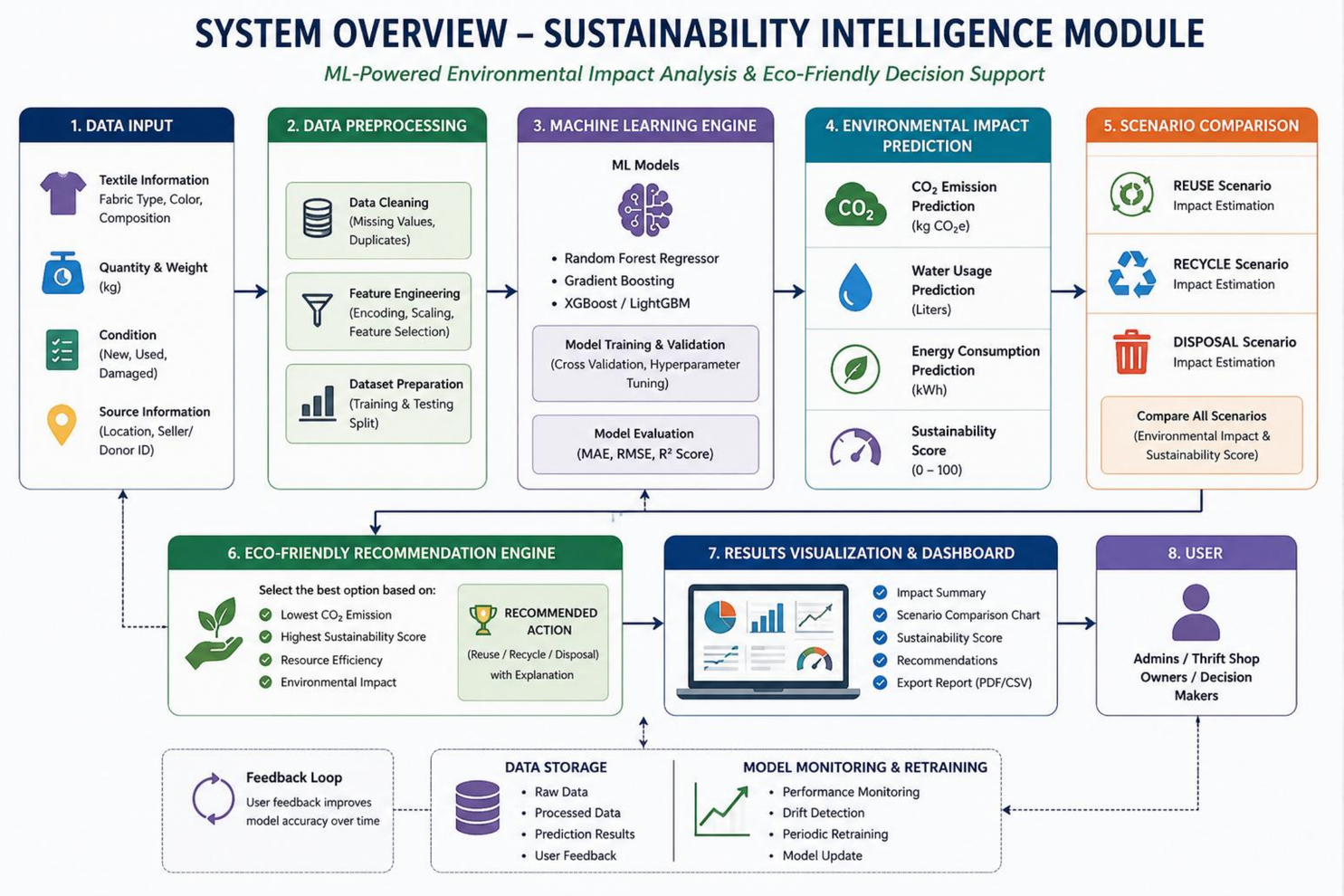


Next Step

- Collect and integrate real textile waste datasets.
- Improve ML model accuracy and perform advanced error analysis.
- Add authentication and admin authorization.
- Integrate Member 01 image analysis outputs into assessment inputs.
- Add production deployment configuration and monitoring support.
- Connect Member 03 pricing and Member 04 marketplace modules.



System Overview Diagram



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Traceability, Recommendation & AI Pricing Module

INTRODUCTION & PROBLEM

- **Existing Problems**

  - ❖ No standard pricing system for textile waste
  - ❖ Difficult to track material transactions
  - ❖ Limited reuse and recycling recommendations
  - ❖ Lack of transparency in textile waste trading
  - ❖ Market demand changes are not considered

- **Proposed AI Solution**

  An AI-powered Smart Value & Traceability Engine that provides:

  - ❖ Intelligent price prediction
  - ❖ Smart buyer/reuse recommendations
  - ❖ Transaction tracking
  - ❖ Digital material records

Novelty Features

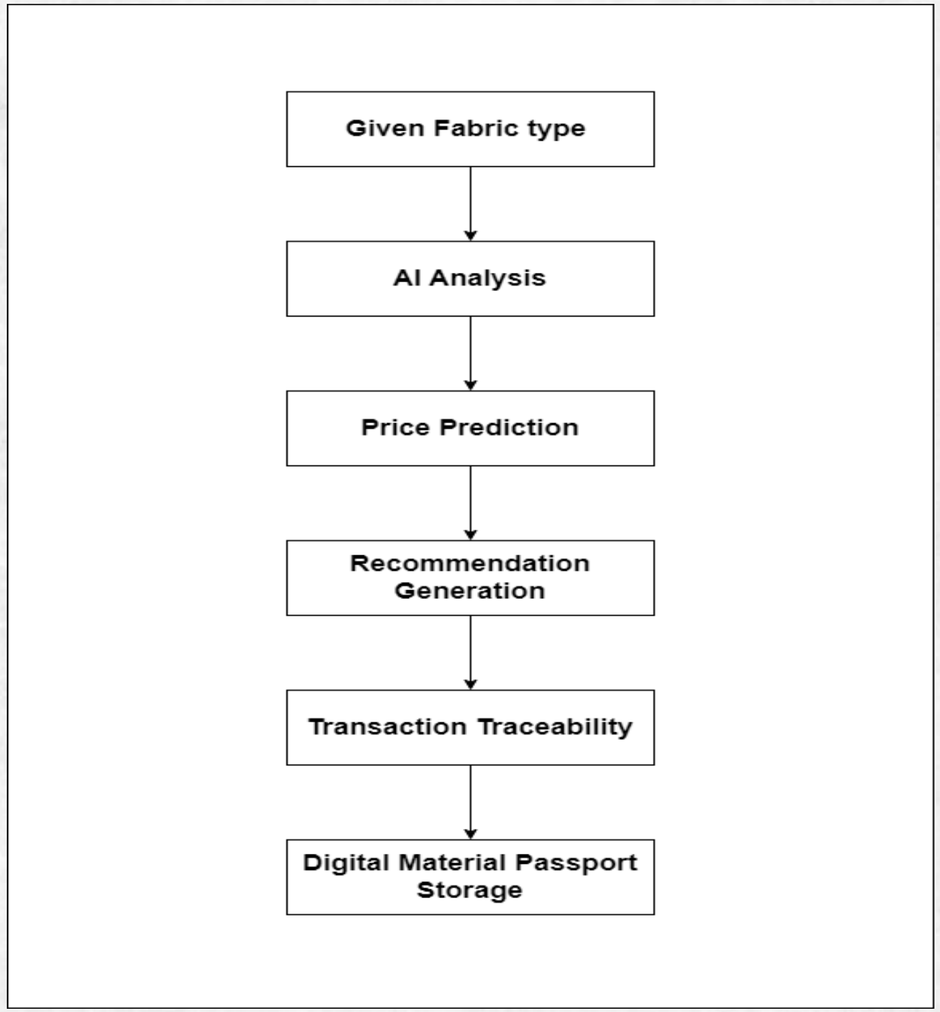
- AI-powered textile waste pricing system
- Demand-aware dynamic price prediction
- Smart recommendation engine for reuse and recycling
- Digital Material Passport for lifecycle tracking
- End-to-end transaction traceability
- Context-aware buyer and reuse recommendations
- Sustainable textile waste management approach
- Data-driven decision support system

Key Features

- AI Pricing Model
- Recommendation System
- Digital Material Passport
- Transaction Traceability
- Demand-aware Pricing
- Context-aware Recommendations
- Buyer & Recycling Suggestions
- Material Lifecycle Tracking

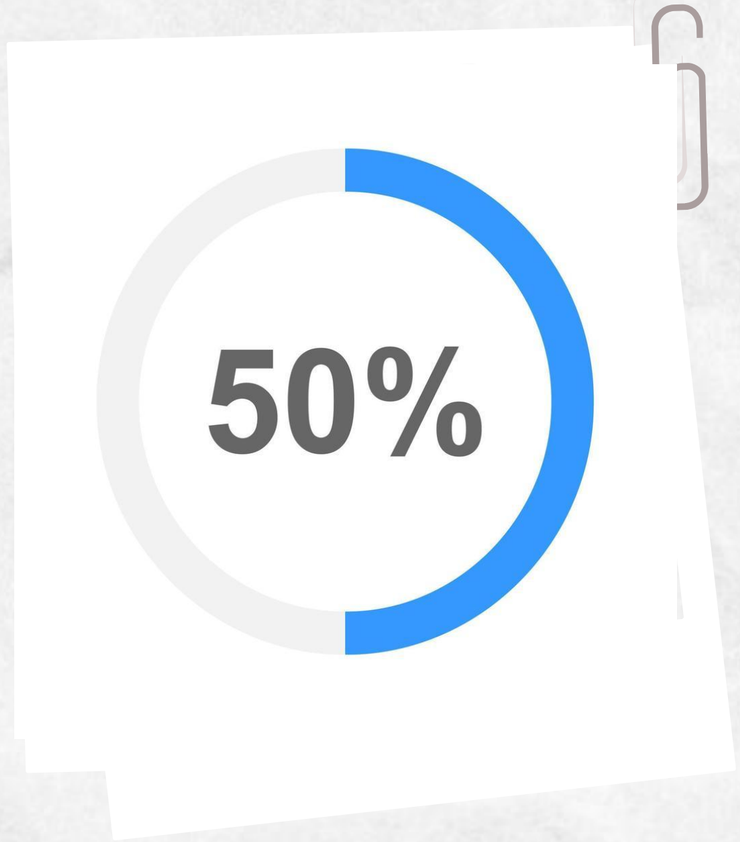
Traceability, Recommendation & AI Pricing Module

System Workflow / Architecture



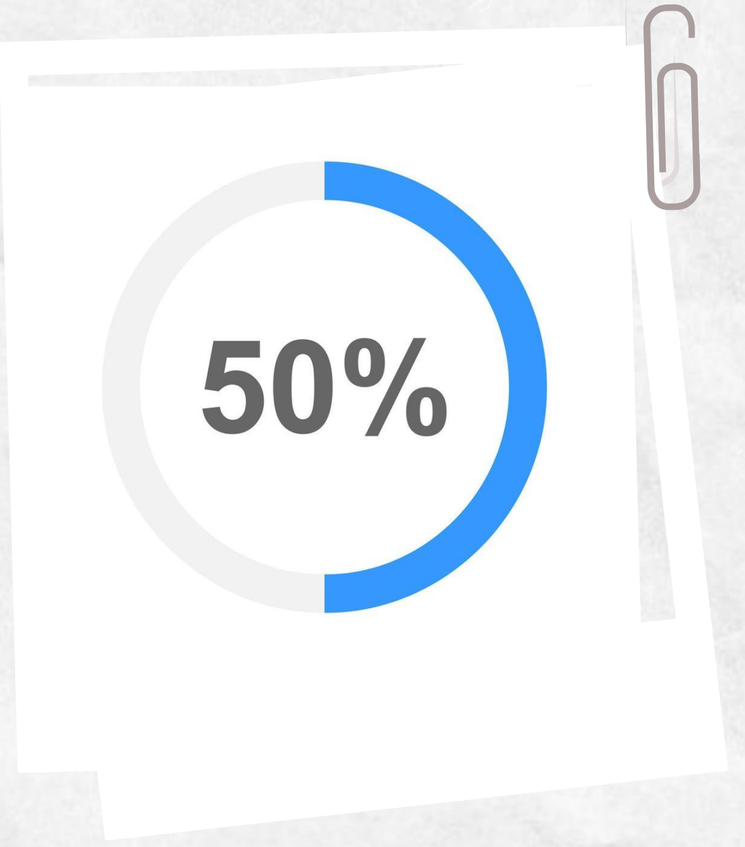
Current Progress

- Dataset collection
- Data preprocessing
- Initial AI model development
- Model training and testing



Next Step

- Recommendation logic design
- UI development
- Backend integration
- Transaction tracking implementation



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Maraketplace & Optimization

Introduction & Problem

What's the problem

Currently, waste management decisions are mostly manual, which leads to:

- inaccurate forecasting,
- inefficient waste handling,
- poor sustainability planning,
- and increased environmental pollution.

Research Gap

- Limited AI-based textile waste forecasting systems
- Lack of material-wise waste prediction
- Absence of real-time analytics dashboards
- Limited integration of AI with waste management
- Poor sustainability decision support
- Limited research focused on Sri Lankan textile waste
- Existing systems focus mainly on historical reporting instead of future forecasting

Evaluation and Validation

- **Model accuracy testing** using labeled textile waste datasets containing material-wise waste quantity records.

- Cross-validation and performance evaluation using machine learning metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

- Comparison of predicted textile waste quantities with historical waste records to evaluate forecasting accuracy and model reliability.

- Validation of material-wise waste forecasting performance for textile materials such as cotton, denim, polyester, nylon, linen, and viscose.

- Analysis of forecasting trends and prediction consistency using historical textile waste data.

Data Availability & Ethical Clearance

- Dataset contains material-wise textile waste quantity records
- No personal or sensitive user data was collected
- Data used only for research and educational purposes
- System follows responsible AI and sustainability-focused practices
- Future work includes larger real-world textile waste datasets and improved forecasting validation

Thank You!